

Deliverable 4: Demonstration Evidence & Execution Logs

Project: Secure Local Encrypted Password Vault Simulator | **Format:** Terminal Execution Evidence

1. Startup & Vault Creation (FR1)

Demonstration of vault creation with interactive master password strength analysis and confirmation:

```
$ python main.py
=====
      SECURE ENCRYPTED PASSWORD VAULT SIMULATOR
=====
      SECURE PASSWORD VAULT (LOCKED)
=====
      [1] Unlock Vault (Log in with Master Password)
      [2] Create a New Vault
      [3] Generate a Strong Random Password
      [4] Exit
-----
Enter your choice or command: 2

CREATE A NEW ENCRYPTED VAULT
-----
Please choose a strong Master Password to protect your vault.
Enter master password: [hidden]
Password Strength: Very Strong (Score: 80/100, Entropy: 86.44 bits)
Confirm master password: [hidden]

Vault created successfully.
```

2. Adding Credential (FR3) & Listing Services (FR4)

```
Enter your choice or command: 1
ADD A NEW CREDENTIAL
-----
Website / App Name: github
Username or Email: user123
Password (hidden while typing): [hidden]

Credential stored successfully.

Enter your choice or command: 3
Stored Services:
-----
1. github
2. gmail
3. linkedin
```

3. Retrieving Credential (FR5) & Explicit Reveal

```
Enter your choice or command: 2
Stored Accounts:
  [1] github (User: user123)

Enter Account Name (e.g. github) or Number: 1

Service: github
Username: user123
Password: *****

Enter your choice or command: 7
Enter Account Name (e.g. github) or Number to reveal password: 1
Are you sure you want to reveal the password for 'github'? [y/N]: y

Password for github: secretpassword123!
```

4. Account Search (FR8), Update (FR6) & Delete (FR7)

```
Enter your choice or command: 6
Enter search keyword (e.g. 'git', 'mail', 'work'): git
Matching services (1 found):
-----
- github (User: user123)

Enter your choice or command: 4
Updating 'github' (press Enter to keep current value):
Username [user123]: [Enter]
New password: [hidden]
Credential updated successfully.

Enter your choice or command: 5
Confirm deletion of 'github'? [y/N]: y
Credential deleted successfully.
```

5. Session Lock (FR9), Brute-Force Rate Limiting & Tamper Detection

```
Enter your choice or command: 12
Vault locked.

Enter your choice or command: GET github
Vault is locked.
Authentication required.

[Failed attempt 5...]
Authentication failed 5 times. Vault is locked for 30 seconds.

[Tampered File Test...]
Vault integrity verification failed! Ciphertext is corrupted or tampered with.
```

6. Plaintext Inspection Security Proof

Inspecting the raw encrypted vault file on disk (data/vault.enc) confirms zero plaintext leakage:

```
$ cat data/vault.enc
{
  "version": 1,
  "kdf": {
    "algorithm": "Argon2id",
    "salt": "l8jG9K+...",
    "params": { "algorithm": "Argon2id", "length": 32, "time_cost": 2, "memory_cost": 65536, "parallelism": 2 }
  },
  "encryption": { "algorithm": "AES-256-GCM" },
  "vault": {
    "nonce": "A8kL+...",
    "ciphertext": "8fA29x92qQ6v8nL..."
  },
  "metadata": { "created_at": "2026-08-19T13:20:00Z", "last_modified": "2026-08-19T13:20:00Z" }
}
```